Introduction to Adam Optimizer

Shuo Guo

August 6, 2026

Contents

Chapter 1

Introduction

Chapter 2

General Setup and Technology Tools

2.1 Adam

Definition 2.1.1 (Innovation).
Let $\mathcal{U}$ be a measurable space. A pair (X, U) consisting of

(i) a $\mathcal{U}$-valued random variable U and

(ii) a product measurable mapping $X\colon \mathcal{U} \times \mathbb{R}^d \to \mathbb{R}^d$

is called innovation.

$\mathcal{U}$ is the random samples. And X puts $\mathcal{U}$ into stochastic gradient.

Definition 2.1.2 (Adam optimizer).
Let

(i) $\alpha, \beta \in [0,1)$, $\epsilon \in (0,\infty)$ *with* $\alpha < \sqrt{\beta}$ *(the* damping parameters*),*

(ii) $n_0 \in \mathbb{N}_0$, $\theta_{n_0}, m_{n_0} \in \mathbb{R}^d$, $v_{n_0} \in [0,\infty)^d$ *(the* initialisation*),*

(iii) a decreasing $(0,\infty)$*-valued sequence* $(\gamma_n)_{n\in\mathbb{N}}$ *(the* sequence of step-sizes*) and*

(iv) (X, U) *an innovation.*

An $\mathbb{R}^d$*-valued stochastic process* $(\theta_n)_{n\in\mathbb{N}_0\cap[n_0,\infty)}$ *is called* Adam algorithm with damping parameters (α,β,ϵ) and step-sizes (γ_n) started at time n_0 in $(\theta_{n_0}, m_{n_0}, v_{n_0})$, *if it satisfies for every* $n \in \mathbb{N}\cap(n_0,\infty)$ *and* $i = 1,\dots,d$

$$\theta_n^{(i)} = \theta_{n-1}^{(i)} + \gamma_n\, \sigma_n^{(i)}\, m_n^{(i)}, \tag{2.1}$$

where $(U_n)_{n\in\mathbb{N}\cap(n_0,\infty)}$ *is a family of independent copies of* U *and for* $n \in \mathbb{N}\cap(n_0,\infty)$ *and* $i = 1,\dots,d$,

- $m_n = \alpha\, m_{n-1} + (1-\alpha)\, X(U_n, \theta_{n-1})$,
- $v_n^{(i)} = \beta\, v_{n-1}^{(i)} + (1-\beta)\,(X^{(i)}(U_n,\theta_{n-1}))^2$ *and*
- $\sigma_n^{(i)} = \dfrac{1}{\sqrt{v_n^{(i)}/(1-\beta^n)} + \epsilon}$.

We call the sequence $(t_n)_{n\in\mathbb{N}_0}$ *given by*

$$t_n = \sum_{k=1}^{n} \gamma_k \tag{2.2}$$

the training times *of the Adam algorithm.*

2.2 Adam sequence space

Consider a sequence of history:

$$\mathbf{x} = (x_k)_{k\in-\mathbb{N}_0} = (x_0, x_{-1}, x_{-2}, \cdots)$$

Consider an operator: $\boxed{\text{The whole history sequence}} \Rightarrow \boxed{\text{Adam direction}}$

Definition 2.2.1.
Let $\alpha, \beta, \epsilon \in [0, \infty)$ *with* $\alpha < \sqrt{\beta} < 1$ *and let* $(\varrho_k)_{k\in-\mathbb{N}_0}$ *as in* (2.4).

$$g\colon \ell^d_\varrho \to \mathbb{R}^d, \qquad \mathbf{x} = (x_k)_{k\in-\mathbb{N}_0} \mapsto \left(\frac{(1-\alpha)\sum_{k\in-\mathbb{N}_0} \alpha^{-k} x_k^{(i)}}{\epsilon + \sqrt{(1-\beta)\sum_{k\in-\mathbb{N}_0} \beta^{-k} (x_k^{(i)})^2}} \right)_{i\in\{1,\ldots,d\}} \tag{2.3}$$

One more natural way is:

$$g(\mathbf{x}) = \frac{m(\mathbf{x})}{\sqrt{v(\mathbf{x})} + \epsilon}$$

We are curious about how *senstive* g is to the change of x_k in $\mathbf{x}$.
One natural idea is to find the partial derivative of g:

$$\begin{aligned} \partial_{x_k} &= \frac{(1-\alpha)\alpha^{-k}}{\sqrt{v(\mathbf{x})} + \epsilon} - \frac{(1-\beta)\beta^{-k} m(\mathbf{x}) x_k}{(\sqrt{v(\mathbf{x})} + \epsilon)^2 \sqrt{v(\mathbf{x})}} \\ &\leq (1-\alpha)\epsilon^{-1}(\alpha^{-k} + \frac{1}{\sqrt{1-\alpha^2/\beta}} \beta^{-\frac{k}{2}}) \end{aligned}$$

So

$$|g(\mathbf{x}) - g(\mathbf{y})| \leq \sum_{k\in-\mathbb{N}_0} (1-\alpha)\epsilon^{-1}(\alpha^{-k} + \frac{1}{\sqrt{1-\alpha^2/\beta}} \beta^{-\frac{k}{2}}) x_k$$

Thus, we can naturally define the norm:

Definition 2.2.2 (Adam sequence space).
Let $\alpha, \beta \in [0, 1)$, $\epsilon \in (0, \infty)$ *with* $\alpha < \sqrt{\beta}$ *be damping factors. We let*

$$\varrho_k = (1-\alpha)\epsilon^{-1}\Big(\alpha^{-k} + \frac{1}{\sqrt{1-\alpha^2/\beta}} \beta^{-k/2}\Big), \qquad \text{for all } k \in -\mathbb{N}_0, \tag{2.4}$$

and denote by ℓ^d_ϱ *the space of all sequences* $\mathbf{x} = (x_k)_{k\in-\mathbb{N}_0} \in (\mathbb{R}^d)^{-\mathbb{N}_0}$ *with*

$$\|\mathbf{x}\|_{\ell^d_\varrho} := \sum_{k\in-\mathbb{N}_0} \rho_k |x_k| < \infty. \tag{2.5}$$

We equip the space with the norm $\|\cdot\|_{\ell_\varrho^d}$. Moreover, we let for a tuple $(m,v)\in\mathbb{R}^d\times[0,\infty)^d$

$$(m,v)_{\ell_\varrho^d} = \inf\Big\{\|\mathbf{x}\|_{\ell_\varrho^d} : \mathbf{x}\in\ell_\varrho^d,\ m=(1-\alpha)\sum_{k\in-\mathbb{N}_0}\alpha^{-k}x_k \ \text{ and } \\ v^{(i)}=(1-\beta)\sum_{k\in-\mathbb{N}_0}\beta^{-k}(x_k^{(i)})^2 \text{ for } i=1,\dots,d\Big\}, \tag{2.6}$$

where the infimum of the empty set is ∞.

Lemma 2.2.3. *Let $\alpha,\beta,\epsilon\in[0,\infty)$ with $\alpha<\sqrt{\beta}<1$ and let $(\varrho_k)_{k\in-\mathbb{N}_0}$ and g as defined in (2.4) and (2.3), respectively. Then g is Lipschitz continuous with Lipschitz constant 1 and g is uniformly bounded by $\sqrt{d}\frac{1-\alpha}{\sqrt{1-\beta}}\frac{1}{\sqrt{1-\alpha^2/\beta}}$.*

Lemma 2.2.4. *For every $k\in-\mathbb{N}_0$, one has that $\varrho_{k-1}\le\sqrt{\beta}\varrho_k$ and the translation operator*

$$T\colon\ell_\varrho^d\to\ell_\varrho^d,\qquad \mathbf{x}=(x_k)_{k\in-\mathbb{N}_0}\mapsto(\mathbb{1}_{\{k\neq0\}}x_{k+1})_{k\in-\mathbb{N}_0} \tag{2.7}$$

is Lipschitz continuous with Lipschitz constant $\sqrt{\beta}<1$. In particular, it maps ℓ_ϱ to ℓ_ϱ.

2.2.1 Proof of Lemma 2.2.3

Proof. We first assume that $d=1$ and briefly write $\ell_\varrho=\ell_\varrho^1$ and $\|\cdot\|_{\ell_\varrho}=\|\cdot\|_{\ell_\varrho^1}$. For $\mathbf{x}\in\ell_\varrho$, let

$$m(\mathbf{x})=(1-\alpha)\sum_{k\in-\mathbb{N}_0}\alpha^{-k}x_k \quad\text{and}\quad v(\mathbf{x})=(1-\beta)\sum_{k\in-\mathbb{N}_0}\beta^{-k}x_k^2 \tag{2.8}$$

so that

$$g(\mathbf{x})=\frac{m(\mathbf{x})}{\sqrt{v(\mathbf{x})}+\epsilon}. \tag{2.9}$$

Using Cauchy-Schwarz and the fact that $\alpha<\sqrt{\beta}$ we get that for all $\mathbf{x}\in\ell_\varrho$

$$\begin{aligned}|m(\mathbf{x})| &= \Big|(1-\alpha)\sum_{k=-\infty}^{0}(\alpha/\sqrt{\beta})^{-k}\sqrt{\beta}^{-k}x_k\Big| \le (1-\alpha)\Big(\sum_{k=0}^{\infty}(\alpha/\sqrt{\beta})^{2k}\Big)^{1/2}\Big(\sum_{k=-\infty}^{0}\beta^{-k}x_k^2\Big)^{1/2}\\ &\le \frac{1-\alpha}{\sqrt{1-\beta}}\frac{1}{\sqrt{1-\alpha^2/\beta}}\sqrt{v(\mathbf{x})}\end{aligned} \tag{2.10}$$

which entails the uniform bound for g that is stated in the lemma for $d=1$. The multivariate estimates follows immediately by applying this estimate onto the individual components.

To prove the Lipschitz continuity we again focus at first on the case $d = 1$. Fix $n \in \mathbb{N}$ and let $\mathbf{x} = (x_0, \dots, x_{-n}, 0, \dots) \in \ell_\varrho$. One has for all $k = 0, \dots, -n$ that

$$\partial_{x_k} g(\mathbf{x}) = (1-\alpha)\alpha^{-k} \frac{1}{\sqrt{v(\mathbf{x})} + \epsilon} - (1-\beta)\beta^{-k} \frac{m(\mathbf{x})}{(\sqrt{v(\mathbf{x})} + \epsilon)^2} \frac{1}{\sqrt{v(\mathbf{x})}} x_k \tag{2.11}$$

so that by using (2.10) and $\sqrt{v(\mathbf{x})} \geq \sqrt{1-\beta}\beta^{-k/2}|x_k|$ we get that

$$|\partial_{x_k} g(\mathbf{x})| \leq (1-\alpha)\epsilon^{-1}\alpha^{-k} + \frac{1-\alpha}{\sqrt{1-\alpha^2/\beta}} \epsilon^{-1}\beta^{-k/2} = \varrho_k. \tag{2.12}$$

For another $\tilde{\mathbf{x}} = (\tilde{x}_0, \dots, \tilde{x}_{-n}, 0, \dots) \in \ell_\varrho$ we let, for $\ell = 0, \dots, n+1$,

$$\mathbf{x}^\ell = (\tilde{x}_0, \dots, \tilde{x}_{-\ell+1}, x_{-\ell}, \dots, x_{-n}, 0, \dots). \tag{2.13}$$

Note that $\mathbf{x}^0 = \mathbf{x}$ and $\mathbf{x}^{n+1} = \tilde{\mathbf{x}}$ and we get with the triangle inequality and the mean value theorem that

$$|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq \sum_{\ell=0}^{n} |g(\mathbf{x}^\ell) - g(\mathbf{x}^{\ell+1})| \leq \sum_{\ell=0}^{n} \varrho_{-\ell}|x_{-\ell} - \tilde{x}_{-\ell}| = \|\mathbf{x} - \tilde{\mathbf{x}}\|_{\ell_\varrho}. \tag{2.14}$$

This inequality remains true for general $\mathbf{x}, \tilde{\mathbf{x}} \in \ell_\varrho$ since

$$g(\mathbf{x}) = \lim_{n\to\infty} g(x_0, \dots, x_{-n}, 0, \dots). \tag{2.15}$$

Indeed, this entails that

$$|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq \limsup_{n\to\infty} |g(x_0, \dots, x_{-n}, 0, \dots) - g(\tilde{x}_0, \dots, \tilde{x}_{-n}, 0, \dots)| \leq \|\mathbf{x} - \tilde{\mathbf{x}}\|_{\ell_\varrho}. \tag{2.16}$$

To obtain the result in the multivariate setting we proceed as above. Fix $n \in \mathbb{N}$, $\mathbf{x} = (x_0, \dots, x_{-n}, 0, \dots) \in \ell_\varrho^d$ and $\tilde{\mathbf{x}} = (\tilde{x}_0, \dots, \tilde{x}_{-n}, 0, \dots) \in \ell_\varrho^d$ and define for every $\ell = 0, \dots, n+1$, $\mathbf{x}^\ell \in \ell_\varrho^d$ accordingly. Then one has for every $\ell = 0, \dots, n$ that

$$|g(\mathbf{x}^\ell) - g(\mathbf{x}^{\ell+1})| = \Big(\sum_{i=1}^{d} |g(\mathbf{x}^{\ell,(i)}) - g(\mathbf{x}^{\ell+1,(i)})|^2\Big)^{1/2} \leq \varrho_{-\ell}\Big(\sum_{i=1}^{d} |x_\ell^{(i)} - \tilde{x}_\ell^{(i)}|^2\Big)^{1/2}, \tag{2.17}$$

where we applied (2.12) for each of the d components. This entails the multivariate variant of estimate (2.14) that is

$$|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq \|\mathbf{x} - \tilde{\mathbf{x}}\|_{\ell_\varrho^d}. \tag{2.18}$$

The rest follows in complete analogy to the one dimensional setting. □

2.2.2 Proof of Lemma 2.2.4

Proof. By definition of (ϱ_k) it immediately follows that, for every $k \in -\mathbb{N}_0$,

$$\varrho_{k-1} \leq \sqrt{\beta}\varrho_k. \tag{2.19}$$

Consequently, one has that, for every $\mathbf{x} \in \ell_\varrho^d$,

$$\|T(\mathbf{x})\|_{\ell_\varrho^d} = \sum_{k\in-\mathbb{N}_0} \varrho_{k-1}|x_k| \leq \sqrt{\beta} \sum_{k\in-\mathbb{N}_0} \varrho_k|x_k| = \sqrt{\beta}\,\|\mathbf{x}\|_{\ell_\varrho^d}. \tag{2.20}$$

The statement follows by linearity of T. □

2.3 Adam vector field

Definition 2.3.1 (Adam vector field).
For a tuple $(\alpha, \beta, \epsilon)$ *of damping parameters and an innovation* (X, U) *we call the mapping* $f : \mathbb{R}^d \to \mathbb{R}^d$ *given by*

$$f^{(i)}(\theta) = (1-\alpha)\,\mathbb{E}^{\theta}\Big[1/\Big(\sqrt{(1-\beta)\sum_{k\in-\mathbb{N}_0}\beta^{-k}X^{(i)}(U_k,\theta)^2}+\epsilon\Big)\sum_{k\in-\mathbb{N}_0}\alpha^{-k}X^{(i)}(U_k,\theta)\Big] \tag{2.21}$$

with $(U_k)_{k\in-\mathbb{N}_0}$ *being a family of independent copies of* U*, the* Adam vector field of the innovation (X, U) for the damping parameters $(\alpha, \beta, \epsilon)$.

Chapter 3

Analysis of Adam evolution and approximation methods

3.1 Analysis of Adam with fixed θ

We study the stochastic optimization algorithm Adam introduced in [?]; its convergence behaviour for general step-size sequences is analysed in [?]. Our analysis follows the classical stochastic approximation framework of [?].

We consider how Adam get update-data(2.1.1 and 2.1.2):

$$X_n = X(U_n, \theta_{n-1})$$

However, it's very hard to give an analysis with such *difficult* setup. We could first fix θ, and see how Adam behaves:

Fix $\theta \in \mathbb{R}^d$ and let $(U_k)_{k\in\mathbb{Z}}$ be an i.i.d. sequence of copies of U. Moreover, for every $n \in \mathbb{Z}$, let $X_n = X(U_n, \theta)$ and $\mathbf{X}(n) = (X_{n+k})_{k\in-\mathbb{N}_0}$. We fix $n_0 \in \mathbb{N}_0$ and consider the process $(\bar{\Theta}_n)_{n\geq n_0}$ given by

$$\bar{\Theta}_n = \sum_{k=n_0+1}^{n} \gamma_k\, g(\mathbf{X}(k)). \tag{3.1}$$

Thus, only $\mathbf{X}$ generated in θ and the stochastic term will drive Adam.

Let Adam walk from $n_0 + 1$ to n. Lemma 3.1.1 will show how *far* $\bar{\Theta}_n$ will be away from it's expection. And Lemma 3.1.2 will analysis the max *error* of the whole path.

Lemma 3.1.1. *Let $p \in [2, \infty)$. One has for every $n \in \mathbb{N}_0$ with $n \geq n_0$ that*

$$\mathbb{E}\big[|\bar{\Theta}_n - \mathbb{E}[\bar{\Theta}_n]|^p\big]^{2/p} \leq \kappa^2 \sum_{k=n_0+1}^{n} \gamma_k^2\, \mathbb{E}\big[|X(U,\theta) - \mathbb{E}[X(U,\theta)]|^p\big]^{2/p}, \tag{3.2}$$

where

$$\kappa = 2C_p\Big((1-\sqrt{\beta})^{-2}\varrho_0\|\varrho\|_{\ell_1} + \|\varrho\|_{\ell_1}^2\Big), \tag{3.3}$$

ϱ is as in (2.4) and C_p is the constant appearing in the the Burkholder-Davis-Gundy inequality for the p-th moment.

Lemma 3.1.2. *Let $p \in (2, \infty)$. One has for every $n \in \mathbb{N}_0$ with $n \geq n_0$*

$$\mathbb{E}\Big[\max_{k=n_0,\dots,n} |\bar{\Theta}_k - \mathbb{E}[\bar{\Theta}_k]|^p\Big]^{1/p} \leq (1 - 2^{-(\frac{1}{2}-\frac{1}{p})})^{-1} \kappa \gamma_{n_0+1} \sqrt{n - n_0}\, \mathbb{E}\Big[|X_0 - \mathbb{E}[X_0]|^p\Big]^{1/p}, \quad (3.4)$$

where κ is as in (3.3).

3.1.1 Proof of Lemma 3.1.1

Our goal is to give an alalysis of

$$\bar{\Theta}_n - \mathbb{E}[\bar{\Theta}_n] \quad (3.5)$$

and we wish to have some form like:

$$\big\|\bar{\Theta}_n - \mathbb{E}[\bar{\Theta}_n]\big\|_{L^p} \lesssim \left(\sum_{k=n_0+1}^{n} \gamma_k^2\right)^{1/2} \|X - \mathbb{E}X\|_{L^p}$$

(below we will analysis why $\left(\sum_{k=n_0+1}^{n} \gamma_k^2\right)^{1/2}$ is a natural estimation)

Doob Martingale Decomposition
We would consider the Doob Martingale and a stopping time N:

$$M_n = \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N] \quad (3.6)$$

As $\mathcal{F}_n$ *increases* (or say: $\cdots, X_{n-1}, X_n.X_{n+1}, \cdots$ get known *one by one*), we could analysis how each term influence the total error estimation:

$$\begin{aligned}
\Delta M_n &= M_n - M_{n-1} \\
&= \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N | \mathcal{F}_{n-1}] \\
&= \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_n\Big] - \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r)) | \mathcal{F}_{n-1}\Big].
\end{aligned}$$

Proof. For ease of notation we prove the statement for $d = 1$. The generalisation to general d is straight-forward. Let $\tilde{X}_0$ be identically distributed as X_0 and independent of the sequence $(X_n)_{n \in \mathbb{Z}}$ and set $\tilde{X}_n = X_n$ for $n \in \mathbb{Z} \backslash \{0\}$. We let

$$\tilde{\mathbf{X}}(n) = (\tilde{X}_{n+k})_{k \in -\mathbb{N}_0}. \quad (3.7)$$

Thus $\tilde{\mathbf{X}}(n)$ agrees with $\mathbf{X}(n)$ in all but at most one component.

Why we need to change $\mathbf{X}(n)$ to $\tilde{\mathbf{X}}(n)$?
To extimate ΔM_n, we are just asking

$$\boxed{\text{How big the inluence of the } n\text{-th random variable } X_n \text{ is to the final result}}$$

So $\tilde{\mathbf{X}}(n)$ agrees with $\mathbf{X}(n)$ in all but at most one component.

This is the thought of standrand replacement (or *Efron-Stein*):

$$\boxed{\text{The importance of a random variable}}$$

$$\Downarrow$$

$$\boxed{\text{change it to an independent copy, how the output change}}$$

We let $(\mathcal{F}_n)_{n\in\mathbb{Z}}$ be the filtration generated by (X_n) and consider for fixed $N \geq n_0$ the martingale $(M_n)_{n\in\mathbb{Z}}$ given by

$$M_n = \mathbb{E}[\bar{\Theta}_N|\mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N] \tag{3.8}$$

We denote by $([M]_n)_{n\in\mathbb{Z}}$ its quadratic variation process being defined by

$$[M]_n = \sum_{r=-\infty}^{n} (M_r - M_{r-1})^2 \tag{3.9}$$

and analyse the $q/2$-th moment of its increments $\Delta[M]_n := (M_n - M_{n-1})^2$. Note that for every $n \in \mathbb{Z} \cap (-\infty, N]$, one has that

$$\Delta M_n = \mathbb{E}[\bar{\Theta}_N|\mathcal{F}_n] - \mathbb{E}[\bar{\Theta}_N|\mathcal{F}_{n-1}] = \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r))|\mathcal{F}_n\Big] - \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r))|\mathcal{F}_{n-1}\Big]. \tag{3.10}$$

Now using that (X_n) is an i.i.d. sequence we conclude that, for almost all $\omega \in \Omega$,

$$\mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r))\Big|\mathcal{F}_n\Big](\omega) = \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r-n))\Big|\forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)\Big] \tag{3.11}$$

and

$$\begin{aligned}\mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r))\Big|\mathcal{F}_{n-1}\Big](\omega) &= \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r-n))\Big|\forall k \in -\mathbb{N} : X_k = X_{k+n}(\omega)\Big] \\ &= \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\tilde{\mathbf{X}}(r-n))\Big|\forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)\Big].\end{aligned} \tag{3.12}$$

Consequently,

$$\begin{aligned}&\mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r))\Big|\mathcal{F}_n\Big](\omega) - \mathbb{E}\Big[\sum_{r=n_0+1}^{N} \gamma_r g(\mathbf{X}(r))\Big|\mathcal{F}_{n-1}\Big](\omega) \\ &\quad = \mathbb{E}\Big[\underbrace{\sum_{r=n_0+1}^{N} \gamma_r (g(\mathbf{X}(r-n)) - g(\tilde{\mathbf{X}}(r-n)))}_{=:\Upsilon_n[N]}\Big|\forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)\Big].\end{aligned} \tag{3.13}$$

We define the totaly inlunce of changing X_0 to $\tilde{X}_0$

$$\Upsilon_n[N] = \sum_{r=n_0+1}^{N} \gamma_r(g(\mathbf{X}(r-n)) - g(\tilde{\mathbf{X}}(r-n))) \tag{3.14}$$

Why we sum for r? Becausr Adam has memory, and X_n will come to several future history:

$$\mathbf{X}_0, \mathbf{X}_1, \cdots$$

Then by *Conditional Expection* and *Jensen*:

$$||\Delta M_n||_{L^p} \leq ||\Upsilon_n[N]||_{L^p}$$

By stationarity, the distribution of the conditional expectation

$$\omega \mapsto \mathbb{E}\big[\Upsilon_n[N]\big|\forall k \in -\mathbb{N}_0 : X_k = X_{k+n}(\omega)\big] \tag{3.15}$$

is the same as the distribution of $\mathbb{E}\big[\Upsilon_n[N]\big|X_0, X_{-1}, \ldots\big]$. Using this and Jensen's inequality we get that

$$\mathbb{E}\big[\Delta[M]_n^{p/2}\big] = \mathbb{E}\big[|M_n - M_{n-1}|^p\big] = \mathbb{E}\big[\big|\mathbb{E}\big[\Upsilon_n[N]\big|X_0, X_{-1}, \ldots\big]\big|^p\big] \leq \mathbb{E}\big[|\Upsilon_n[N]|^p\big]. \tag{3.16}$$

It follows with the triangle inequality that

$$\mathbb{E}\big[[M]_N^{p/2}\big]^{2/p} = \mathbb{E}\Big[\Big(\sum_{n=-\infty}^{N} \Delta[M]_n\Big)^{p/2}\Big]^{2/p} \leq \sum_{n=-\infty}^{N} \mathbb{E}\big[(\Delta[M]_n)^{p/2}\big]^{2/p} \leq \sum_{n=-\infty}^{N} \mathbb{E}\big[|\Upsilon_n[N]|^p\big]^{2/p}. \tag{3.17}$$

ϱ_{n-r}: How old X_n is in r-th ipdate.
$|X_0 - \tilde{X}_0|$: the origin error of replacing the random sample.

Define the scale of the noise:

$$C := \mathbb{E}[|X_0 - \mathbb{E}[X_0]|^p]^{1/p}$$

As X_0 are independent to $\tilde{X}_0$, we have

$$X_0 - \tilde{X}_0 = (X_0 - \mathbb{E}(X_0)) - (\tilde{X}_0 - \mathbb{E}(X_0))$$

By Minkowski inequality

$$||X_0 - \tilde{X}_0||_{L^p} \leq ||X_0 - \mathbb{E}(X_0)||_{L^p} + ||\tilde{X}_0 - \mathbb{E}(X_0)||_{L^p} = 2C$$

Now for $n \in \mathbb{Z}$ with $n \leq N$ we get with Lemma 2.2.3 that

$$|\Upsilon_n[N]| \leq \sum_{r=n\vee(n_0+1)}^{N} \gamma_r|g(\mathbf{X}(r-n)) - g(\tilde{\mathbf{X}}(r-n))| \leq \sum_{r=n\vee(n_0+1)}^{N} \gamma_r \varrho_{n-r}|X_0 - \tilde{X}_0| \tag{3.18}$$

and again by the triangle inequality we have in terms of $C := \mathbb{E}[|X_0 - \mathbb{E}[X_0]|^p]^{1/p}$,

$$\begin{aligned}\mathbb{E}[|\Upsilon_n[N]|^p]^{1/p} &\le \sum_{r=n\vee(n_0+1)}^{N} \gamma_r \varrho_{n-r}\, \mathbb{E}[|X_0 - \tilde{X}_0|^p]^{1/p} \\ &\le \begin{cases} 2C\|\varrho\|_{\ell_1}\gamma_n, & \text{if } n \ge n_0+1, \\ 2C(1-\sqrt{\beta})^{-1}\varrho_{n-(n_0+1)}\gamma_{n_0+1}, & \text{if } n \le n_0. \end{cases}\end{aligned} \tag{3.19}$$

If $n \ge n_0 + 1$: Samples truly newly generated within the optimization interval.

$$\|\Upsilon_n[N]\|_{L^p} \le 2C\varrho_n \sum_{r=n}^{N} \varrho_{n-r} \le 2C\gamma_n\|\varrho\|_{l^1}, \quad \text{as } \{\gamma_n\} \text{ is decreasing}$$

Hence we obtain

$$2C\gamma_n\|\varrho\|_{l^1}, \quad \text{where } \|\varrho\|_{l^1} = \sum_{j\le 0}\varrho_j.$$

If $n \le n_0$ Old historical samples from before the start of optimization.

$$\begin{aligned}\|\Upsilon_n[N]\|_{L^p} &\le 2C\gamma_{n_0+1}\sum_{r=n_0+1}^{N}\varrho_{n-r} \\ &\le 2C\gamma_{n_0+1}\varrho_{n-(n_0+1)}\sum_{j=0}^{\infty}(\sqrt{\beta})^j \\ &\le 2C(1-\sqrt{\beta})^{-1}\varrho_{n-(n_0+1)}\gamma_{n_0+1}.\end{aligned}$$

with

$$\varrho_{k-1} \le \sqrt{\beta}\varrho_k.$$

Together with (3.17) we conclude that

$$\begin{aligned}\mathbb{E}\big[[M]_N^{p/2}\big]^{2/p} &\le \sum_{n=-\infty}^{N}\mathbb{E}[|\Upsilon_n[N]|^p]^{2/p} \\ &\le 4C^2(1-\sqrt{\beta})^{-2}\gamma_{n_0+1}^2\sum_{n=-\infty}^{n_0}\varrho_{n-(n_0+1)}^2 + 4C^2\|\varrho\|_{\ell_1}^2\sum_{n=n_0+1}^{N}\gamma_n^2 \\ &\le 4C^2\Big((1-\sqrt{\beta})^{-2}\varrho_0\|\varrho\|_{\ell_1}\gamma_{n_0+1}^2 + \|\varrho\|_{\ell_1}^2\sum_{n=n_0+1}^{N}\gamma_n^2\Big).\end{aligned} \tag{3.20}$$

The statement follows with the Burkholder-Davis-Gundy inequality since $\bar{\Theta}_N = \mathbb{E}[\bar{\Theta}_N] + M_N$. □

3.1.2 Proof of Lemma 3.1.2

Proof. Without loss of generality we can assume that $n_0 = 0$ since the general statement then follows by an index shift.

Suppose that $m \in \mathbb{N}$ and $2^{m-1} \le N \le 2^m - 1$. We call for every $k = 1, \dots, m$, all intervals of the form

$$[j, j+2^{m-k}) \qquad (j \in (2^{m-k+1}\mathbb{N}_0) \cap \{0, \dots, N\}) \tag{3.21}$$

and additionally the empty set an interval of the k-th layer. Note that by construction for every natural number $n \in \{0, \dots, N\}$ the interval $[0, n)$ can be uniquely written as pairwise disjoint union

$$[0, n) = [0, n_1) \cup [n_1, n_2) \cup \dots \cup [n_{m-1}, n_m) \tag{3.22}$$

with each $[n_{k-1}, n_k)$ being an interval of the k-th layer. Indeed, the representation is obtained for $n_k = 2^{m-k} \lfloor 2^{-(m-k)} n \rfloor$. We let $\bar{\Theta}_{k,\ell} := \bar{\Theta}_\ell - \bar{\Theta}_k - \mathbb{E}[\bar{\Theta}_\ell - \bar{\Theta}_k]$ for every $k, \ell \in \mathbb{N}_0$ with $0 \le k \le \ell$. Then for n as above we have

$$\bar{\Theta}_{0,n} = \sum_{k=1}^{m} \bar{\Theta}_{n_{k-1}, n_k}. \tag{3.23}$$

Consequently,

$$\max_{n \in \{0, \dots, N\}} |\bar{\Theta}_{0,n}| \le \sum_{k=1}^{m} \max_{[n_{k-1}, n_k)\text{: interval of } k\text{th layer}} |\bar{\Theta}_{n_{k-1}, n_k}| \tag{3.24}$$

and

$$\mathbb{E}\Big[\max_{n \in \{0, \dots, N\}} |\bar{\Theta}_{0,n}|^p\Big]^{1/p} \le \sum_{k=1}^{m} \mathbb{E}\Big[\max_{[n_{k-1}, n_k)\text{: interval of } k\text{th layer}} |\bar{\Theta}_{n_{k-1}, n_k}|^p\Big]^{1/p}. \tag{3.25}$$

Now note that by Lemma 3.1.1 one has in terms of $C := \mathbb{E}[|X_0 - \mathbb{E}[X_0]|^p]^{1/p}$ that for an arbitrary non-empty interval $[n_{k-1}, n_k)$ of the k-th layer

$$\mathbb{E}[|\bar{\Theta}_{n_{k-1}:n_k}|^p]^{2/p} \le \kappa^2 \gamma_1^2 2^{m-k} C^2 \tag{3.26}$$

so that

$$\begin{aligned} \mathbb{E}\Big[\max_{[n_{k-1}, n_k)\text{: interval of } k\text{th layer}} |\bar{\Theta}_{n_{k-1}, n_k}|^p\Big] &\le \sum_{[n_{k-1}, n_k)\text{: interval of } k\text{th layer}} \mathbb{E}[|\bar{\Theta}_{n_{k-1}, n_k}|^p] \\ &\le \kappa^p \gamma_1^p C^p 2^{k-1} (2^{m-k})^{p/2}. \end{aligned} \tag{3.27}$$

Together with (3.25) we obtain that

$$\mathbb{E}\Big[\max_{n \in \{0, \dots, N\}} |\bar{\Theta}_{0,n}|^p\Big]^{1/p} \le \kappa \gamma_1 C \sqrt{2^{m-1}} \sum_{k=1}^{m} 2^{-(\frac{1}{2} - \frac{1}{p})(k-1)} \le (1 - 2^{-(\frac{1}{2} - \frac{1}{p})})^{-1} \kappa \gamma_1 C \sqrt{N}. \tag{3.28}$$

□

3.2 Approximation to Adam process

For this we fix damping parameters $\alpha, \beta, \epsilon \in [0, \infty)$ with $\alpha < \sqrt{\beta} < 1$ and an innovation (X, U) and denote by $(\theta_n)_{n \geq n_0}$ an Adam algorithm started at time $n_0 \in \mathbb{N}_0$ in a state $(\theta_{n_0}, m_{n_0}, v_{n_0})$. We assume that there exists a deterministic $\mathbf{x} = (x_k)_{k \in -\mathbb{N}_0} \in \ell^d_\varrho$ such that for all $i \in \{1, \ldots, d\}$ we have

$$m_{n_0} = (1 - \alpha) \sum_{k \in -\mathbb{N}_0} \alpha^{-k} x_k \qquad \text{and} \qquad v^{(i)}_{n_0} = (1 - \beta) \sum_{k \in -\mathbb{N}_0} \beta^{-k} (x^{(i)}_k)^2. \tag{3.29}$$

We consider the $\mathbb{R}^d$-valued process $(X_k)_{k \in \mathbb{Z}}$ given by

$$X_k = \begin{cases} x_{k-n_0}, & \text{if } k \leq n_0, \\ X(U_k, \theta_{k-1}), & \text{if } k > n_0 \end{cases} \tag{3.30}$$

First approximation:

Dropping out the correction term $(1 - \beta^n)$:

$$\Theta_n := \sum_{k=n_0+1}^{n} \gamma_k \, g(\mathbf{X}(k)). \tag{3.31}$$

Second approximation:
From n_0, we cut the future terms with $(n_\ell)_{\ell \in \mathbb{N}_0}$. Thus we have *intervals* like:

$$\{n_0 + 1, n_0 + 2, \cdots, n_1\}$$

$$\{n_1 + 1, n_1 + 2, \cdots, n_2\}$$

$$\vdots$$

For every $\ell \in \mathbb{N}$ and $n \in \{n_{\ell-1} + 1, \ldots, n_\ell\}$, *(understand n as stopping time, though not exactly)* the process $\tilde{\mathbf{X}}(n) = (\tilde{\mathbf{X}}(n)_k)_{k \in -\mathbb{N}_0}$ by

$$\tilde{\mathbf{X}}(n)_k = \begin{cases} X(U_{n+k}, \theta_{n_{\ell-1}}), & \text{if } n + k > n_{\ell-1}, \\ X_{n+k}, & \text{else.} \end{cases} \tag{3.32}$$

Comparing $\mathbf{X}(n)$ and $\tilde{\mathbf{X}}(n)$ we see that the terms outside the $(n_{\ell-1}, n_\ell]$-window agree and inside the window the θ-parameter is fixed as $\theta_{n_{\ell-1}}$. In analogy to before we consider the process $(\tilde{\Theta}_n)_{n \geq n_0}$ given by

$$\tilde{\Theta}_n := \sum_{k=n_0+1}^{n} \gamma_k \, g(\tilde{\mathbf{X}}(k)). \tag{3.33}$$

Third approximation

Next, let $(U_{\ell,n})_{\ell \in \mathbb{N}, n \in \mathbb{Z}}$ be a family of independent copies of U that is also independent of $(U_n)_{n \in \mathbb{N}}$. For every $\ell \in \mathbb{N}$ and $n \in \{n_{\ell-1}+1, \ldots, n_\ell\}$, we consider $\tilde{\tilde{\mathbf{X}}}(n) = (\tilde{\tilde{\mathbf{X}}}(n)_k)_{k \in -\mathbb{N}_0}$

given by

$$\tilde{\tilde{\mathbf{X}}}(n)_k = \begin{cases} X(U_{n+k}, \theta_{n_{\ell-1}}) = \tilde{\mathbf{X}}(n)_k, & \text{if } n+k > n_{\ell-1}, \\ X(U_{\ell,n+k}, \theta_{n_{\ell-1}}), & \text{else.} \end{cases} \tag{3.34}$$

In analogy to before we define a process $(\tilde{\tilde{\Theta}}_n)_{n \geq n_0}$ via

$$\tilde{\tilde{\Theta}}_n = \sum_{k=n_0+1}^{n} \gamma_k\, g(\tilde{\tilde{\mathbf{X}}}(k)). \tag{3.35}$$

Understand the third approximation as re-write the whole history with the location $\theta_{n_{\ell-1}}$.

Then we come to the result of approximation:

Proposition 3.2.1. *Let*

(i) $\alpha, \beta \in [0,1)$ *with* $\alpha < \sqrt{\beta}$ *and* $\epsilon \in (0,\infty)$ *(the* damping parameters*),*

(ii) $n_0 \in \mathbb{N}_0$, $\theta_{n_0}, m_{n_0} \in \mathbb{R}^d$, $v_{n_0} \in [0,\infty)^d$ *(the initialisation),*

(iii) a decreasing $(0,\infty)$*-valued sequence* $(\gamma_n)_{n\in\mathbb{N}}$ *(*sequence of step-sizes*),*

(iv) (X,U) *an innovation,*

(v) $C, \tilde{C}, \tilde{L} \in [0,\infty)$, $p \in [2,\infty)$ *and* $V \subset \mathbb{R}^d$ *be a measurable set,*

(vi) $(n_\ell)_{\ell\in\mathbb{N}_0}$ *be a strictly increasing* $\mathbb{N}_0$*-valued sequence.*

Let (X,U) *be a p-regular innovation with parameter* $(C,\tilde{C},\tilde{L})$ *on* V *and* $(\theta_n)_{n\in\mathbb{N}_0\cap[n_0,\infty)}$ *the Adam algorithm with damping parameter* (α,β,ϵ) *and step-sizes* (γ_n) *started at time* n_0 *in* $(\theta_{n_0}, m_{n_0}, v_{n_0})$*. Moreover, let*

$$\mathfrak{N} = \inf\{n \geq n_0 : \theta(n) \notin V\} \tag{3.36}$$

and let $(t_n)_{n\in\mathbb{N}}$ *as in (2.2). One has the following for the approximations* $(\Theta_n)_{n\geq n_0}$, $(\tilde{\Theta}_n)_{n\geq n_0}$ *and* $(\tilde{\tilde{\Theta}}_n)_{n\geq n_0}$*:*

(I) for every $\mathfrak{n} \in \{n_0, n_0+1, \dots\}$,

$$\mathbb{E}\Big[\Big(\sum_{k=\mathfrak{n}+1}^{\infty} \mathbb{1}_{\{\mathfrak{N}\geq k\}} |\Delta\theta_k - \Delta\Theta_k|\Big)^p\Big]^{1/p} \leq \Big(\kappa_1 \beta^{\frac{1}{2}(\mathfrak{n}+1)-n_0} \|\mathbf{x}\|_{\ell^d_\varrho} + \kappa_2 C\Big)\gamma_{\mathfrak{n}+1}\beta^{\mathfrak{n}+1} \tag{3.37}$$

(II) for every $\ell \in \mathbb{N}$ *and* $\mathfrak{n}, n \in \mathbb{N}_0$ *with* $n_{\ell-1} \leq \mathfrak{n} \leq n \leq n_\ell$,

$$\mathbb{E}\Big[\Big(\sum_{k=\mathfrak{n}+1}^{n} \mathbb{1}_{\{\mathfrak{N}\geq k\}} |\Delta\Theta_k - \Delta\tilde{\Theta}_k|\Big)^p\Big]^{1/p} \leq \kappa_3 \tilde{L}(t_{n_\ell} - t_{n_{\ell-1}})(t_n - t_{\mathfrak{n}}) \tag{3.38}$$

(III) for every $\ell \in \mathbb{N}$,

$$\mathbb{E}\Big[\mathbb{1}_{\{\mathfrak{N}>n_{\ell-1}\}}\Big(\sum_{k=n_{\ell-1}+1}^{n_\ell} |\Delta\tilde{\Theta}_k - \Delta\tilde{\tilde{\Theta}}_k|\Big)^p\Big]^{1/p} \leq \gamma_{n_{\ell-1}+1}\Big(\kappa_4 C + \kappa_5 \beta^{(n_{\ell-1}-n_0)/2} \|\mathbf{x}\|_{\ell^d_\varrho}\Big), \tag{3.39}$$

(IV.a) (Recall Lemma 3.1.1) for every $\ell \in \mathbb{N}$,

$$\mathbb{E}[\mathbb{1}_{\{\mathfrak{N}>n_{\ell-1}\}}|\tilde{\tilde{\Theta}}_{n_\ell} - \tilde{\tilde{\Theta}}_{n_{\ell-1}} - (t_{n_\ell} - t_{n_{\ell-1}})f(\theta_{n_{\ell-1}})|^p|\mathcal{F}_{n_{\ell-1}}]^{1/p} \leq \kappa_6\tilde{C}\Big(\sum_{k=n_{\ell-1}+1}^{n_\ell}\gamma_k^2\Big)^{1/2}, \tag{3.40}$$

(IV.b) (Recall Lemma 3.1.2) if $p > 2$, *then for every* $\ell \in \mathbb{N}$,

$$\begin{aligned}\mathbb{E}[\mathbb{1}_{\{\mathfrak{N}>n_{\ell-1}\}}\max_{n=n_{\ell-1},\dots,n_\ell}|\tilde{\tilde{\Theta}}_n - \tilde{\tilde{\Theta}}_{n_{\ell-1}} - (t_n - t_{n_{\ell-1}})f(\theta_{n_{\ell-1}})|^p|\mathcal{F}_{n_{\ell-1}}]^{1/p} \\ \leq (1 - 2^{-(\frac{1}{2}-\frac{1}{p})})^{-1}\kappa_6\tilde{C}\gamma_{n_{\ell-1}+1}\sqrt{n_\ell - n_{\ell-1}},\end{aligned} \tag{3.41}$$

where

$$\begin{aligned}&\kappa_1 = \frac{2}{1-\beta^{3/2}},\ \kappa_2 = \frac{2}{1-\beta^{3/2}}\Big(\frac{\varrho_0}{1-\beta} + \|\varrho\|_{\ell_1}\Big), \kappa_3 = \frac{1-\alpha}{\sqrt{1-\beta}\sqrt{1-\alpha^2/\beta}}d\|\varrho\|_{\ell_1}, \\ &\kappa_4 = 2(1-\sqrt{\beta})^{-1}\|\varrho\|_{\ell_1},\ \kappa_5 = (1-\sqrt{\beta})^{-1},\ \kappa_6 = 2C_p\Big((1-\sqrt{\beta})^{-1}\sqrt{\varrho_0\|\varrho\|_{\ell_1}} + \|\varrho\|_{\ell_1}\Big),\end{aligned} \tag{3.42}$$

(ϱ_k) *is as in (2.4) and* C_p *is the constant in the Burholder-Davis-Gundy inequality when applied for the pth moment.*

For 3.43,

$$\mathbb{E}\Big[\Big(\sum_{k=\mathfrak{n}+1}^{n}\mathbb{1}_{\{\mathfrak{N}\geq k\}}|\Delta\Theta_k - \Delta\tilde{\Theta}_k|\Big)^p\Big]^{1/p} \leq \kappa_3\tilde{L}(t_{n_\ell} - t_{n_{\ell-1}})(t_n - t_{\mathfrak{n}}) \tag{3.43}$$

Define the ODE time length in every block as

$$h_\ell := t_{n_\ell} - t_{n_{\ell-1}}$$

And sum the error with the learning rate as

$$\sum_{k=n_{\ell-1}+1}^{n_\ell} = h_\ell$$

So the error from Frozen parameter in block is $\mathcal{O}(h_\ell^2)$.

Part	Error source	Magnitude
(I)	finite-time bias correction	$\mathcal{O}(\gamma_n\beta^n)$
(II)	Frozen parameter in block	$\mathcal{O}(h_\ell^2)$
(III)	Old real history replaced by stationary history	$\mathcal{O}(\gamma_\ell^*)$
(IV)	Random fluctuations	$\mathcal{O}\big((\sum_{block}\gamma_K^2)^{1/2}\big)$

Using Part (II) to the block, we can get

$$E_{frozen} = \mathcal{O}(h_\ell^2)$$

Using Part (III) to the block, we can get

$$E_{memory} = \mathcal{O}(\gamma_\ell^*)$$

And we hope that E_{frozen} is comparable with E_{memory}:

$$h_\ell^2 \asymp \gamma_\ell^*$$

Then we have the motibation of definition 3.2.2.

Definition 3.2.2.
*Let $n_0 \in \mathbb{N}_0$ and $\rho \in [\sqrt{\gamma_{n_0+1}}, \infty)$. We call the $\mathbb{N}_0$-valued sequence $(n_\ell)_{\ell\in\mathbb{N}_0}$ a ρ-*partition *w.r.t.* (γ_n) *if for every $\ell \in \mathbb{N}$, n_ℓ is the largest integer with*

$$t_{n_\ell} - t_{n_{\ell-1}} \leq \rho\sqrt{\gamma_{n_{\ell-1}+1}}, \tag{3.44}$$

where (t_n) is as in (2.2). We call n_0 the starting value.

Remark 3.2.3. In the definition we assume that $\rho \geq \sqrt{\gamma_{n_0+1}}$. This guarantees that for every $\ell \in \mathbb{N}$, one has $\rho\sqrt{\gamma_{n_{\ell-1}+1}} \geq \gamma_{n_{\ell-1}+1}$ by monotonicity of (γ_n) so that $n_\ell > n_{\ell-1}$. This implies that (n_ℓ) is strictly increasing and that $\lim_{n\to\infty} n_\ell = \infty$.

Lemma 3.2.4. *Let $(\gamma_n)_{n\in\mathbb{N}}$ be a $(0,\infty)$-valued decreasing non-summable sequence and let $(n_\ell)_{\ell\in\mathbb{N}_0}$ be a ρ-partition for (γ_n).*

(i) For $\ell \in \mathbb{N}$ with $\gamma_{n_{\ell-1}+1} < \rho^2$, one has that

$$\sqrt{\gamma_{n_{\ell-1}+1}} \leq \frac{1}{\rho - \sqrt{\gamma_{n_{\ell-1}+1}}}(t_{n_\ell} - t_{n_{\ell-1}}). \tag{3.45}$$

(ii) Assume that for all $n = n_0+1, n_0+2, \ldots$

$$\frac{\gamma_n - \gamma_{n+1}}{\gamma_n^2} \leq \zeta \tag{3.46}$$

and that $\rho\zeta\sqrt{\gamma_{n_0+1}} < 1$. Set $K = (1-\zeta\rho\sqrt{\gamma_{n_0+1}})^{-1}$. One has for every $\ell \in \mathbb{N}$ that

$$\frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_\ell+1}} \leq K \quad \text{and} \quad \frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_\ell+1}} \leq 1 + \zeta K(t_{n_\ell} - t_{n_{\ell-1}}). \tag{3.47}$$

3.2.1 Proof of Proposition 3.2.1

Proof. 1) We prove the first inequality. Note that

$$\begin{aligned}\left|\sigma_k^{(i)} m_k^{(i)} - g(\mathbf{X}^{(i)}(k))\right| &\leq \left(1 - \frac{1}{\sqrt{1/(1-\beta^k)}}\right)|g(\mathbf{X}^{(i)}(k))| \leq 2\beta^k \|\mathbf{X}^{(i)}(k)\|_{\ell_\varrho} \\ &\leq 2\beta^k\left(\sqrt{\beta}^{k-n_0}\|\mathbf{x}^{(i)}\|_{\ell_\varrho} + \sum_{r=n_0+1}^{k} \varrho_{r-k}|X_r^{(i)}|\right).\end{aligned} \tag{3.48}$$

This entails that

$$\mathbb{E}[\mathbb{1}_{\{\mathfrak{N}\geq k\}}|\Delta\theta_k-\Delta\Theta_k|^p]^{1/p}\leq 2\beta^k\Big(\sqrt{\beta}^{k-n_0}\|\mathbf{x}\|_{\ell^d_\varrho}+C\sum_{r=n_0+1}^{k}\varrho_{r-k}\Big). \tag{3.49}$$

Consequently,

$$\begin{aligned}\mathbb{E}\Big[\Big(\sum_{k=\mathfrak{n}+1}^{\infty}\mathbb{1}_{\{\mathfrak{N}\geq k\}}\gamma_k|\Delta\theta_k-\Delta\Theta_k|\Big)^p\Big]^{1/p}&\leq 2\sum_{k=\mathfrak{n}+1}^{\infty}\gamma_k\beta^k\Big(\sqrt{\beta}^{k-n_0}\|\mathbf{x}\|_{\ell^d_\varrho}+C\sum_{r=n_0+1}^{k}\varrho_{r-k}\Big)\\&\leq 2(1-\beta^{3/2})^{-1}\gamma_{\mathfrak{n}+1}\beta^{\frac{3}{2}(\mathfrak{n}+1)-n_0}\|\mathbf{x}\|_{\ell^d_\varrho}+2C\gamma_{\mathfrak{n}+1}\sum_{r=n_0+1}^{\infty}\sum_{k=r\vee(\mathfrak{n}+1)}^{n}\beta^k\varrho_{r-k}\end{aligned} \tag{3.50}$$

In the case where $r\leq\mathfrak{n}$, one has that

$$\sum_{k=r\vee(\mathfrak{n}+1)}^{n}\beta^k\varrho_{r-k}\leq(1-\beta^{3/2})^{-1}\varrho_{r-(\mathfrak{n}+1)}\beta^{\mathfrak{n}+1} \tag{3.51}$$

and in the case where $r>\mathfrak{n}$,

$$\sum_{k=r\vee(\mathfrak{n}+1)}^{n}\beta^k\varrho_{r-k}\leq(1-\beta^{3/2})^{-1}\varrho_0\beta^r. \tag{3.52}$$

Hence,

$$\begin{aligned}&\mathbb{E}\Big[\Big(\sum_{k=\mathfrak{n}+1}^{\infty}\mathbb{1}_{\{\mathfrak{N}\geq k\}}\gamma_k|\Delta\theta_k-\Delta\Theta_k|\Big)^p\Big]^{1/p}\\&\leq 2(1-\beta^{3/2})^{-1}\gamma_{\mathfrak{n}+1}\Big(\beta^{\frac{3}{2}(\mathfrak{n}+1)-n_0}\|\mathbf{x}\|_{\ell^d_\varrho}+\beta^{\mathfrak{n}+1}C\sum_{r=n_0+1}^{\mathfrak{n}}\varrho_{r-(\mathfrak{n}+1)}+C\sum_{r=\mathfrak{n}+1}^{n}\varrho_0\beta^r\Big)\\&\leq 2(1-\beta^{3/2})^{-1}\Big(\beta^{\frac{1}{2}(\mathfrak{n}+1)-n_0}\|\mathbf{x}\|_{\ell^d_\varrho}+(\|\varrho\|_{\ell_1}+(1-\beta)^{-1}\varrho_0)C\Big)\gamma_{\mathfrak{n}+1}\beta^{\mathfrak{n}+1}.\end{aligned} \tag{3.53}$$

2.) Fix $\ell\in\mathbb{N}$. Since for every $k\geq n_0$

$$\sigma_k^{(i)}m_k^{(i)}\leq\frac{1-\alpha}{\sqrt{1-\beta}}\frac{1}{\sqrt{1-\alpha^2/\beta}}, \tag{3.54}$$

one has for every $n=n_{\ell-1}+1,\dots,n_\ell$

$$|\theta_n^{(i)}-\theta_{n_{\ell-1}}^{(i)}|=\Big|\sum_{k=n_{\ell-1}+1}^{n}\gamma_k\sigma_k^{(i)}m_k^{(i)}\Big|\leq\frac{1-\alpha}{\sqrt{1-\beta}}\frac{1}{\sqrt{1-\alpha^2/\beta}}(t_n-t_{n_{\ell-1}}). \tag{3.55}$$

This implies that

$$\mathbb{E}[\mathbb{1}_{\{\mathfrak{N}\geq n\}}|X(U_n,\theta_{n-1})-X(U_n,\theta_{n_{\ell-1}})|^p|\mathcal{F}_{n-1}]^{1/p}\leq\underbrace{\frac{1-\alpha}{\sqrt{1-\beta}}\frac{1}{\sqrt{1-\alpha^2/\beta}}d}_{=\kappa_3/\|\varrho\|_{\ell_1}}\tilde{L}(t_{n_\ell}-t_{n_{\ell-1}}). \tag{3.56}$$

Now note that for $k=n_{\ell-1}+1,\dots,n_\ell$

$$\big|g(\mathbf{X}(k))-g(\tilde{\mathbf{X}}(k))\big|\leq\sum_{r=n_{\ell-1}+1}^{k}\rho_{r-k}|X(U_r,\theta_{r-1})-X(U_r,\theta_{n_{\ell-1}})|. \tag{3.57}$$

This implies that for $\mathfrak{n}, n \in \mathbb{N}_0$ with $n_{\ell-1} \le \mathfrak{n} \le n \le n_\ell$

$$
\begin{aligned}
&\mathbb{E}\Big[\Big(\sum_{k=\mathfrak{n}+1}^{n} \mathbb{1}_{\{\mathfrak{N}>k\}}\gamma_k\big|g(\mathbf{X}(k)) - g(\tilde{\mathbf{X}}(k))\big|\Big)^p\Big]^{1/p} \\
&\quad\le \sum_{k=\mathfrak{n}+1}^{n} \gamma_k\, \mathbb{E}\Big[\Big(\mathbb{1}_{\{\mathfrak{N}>k\}}\big|g(\mathbf{X}(k)) - g(\tilde{\mathbf{X}}(k))\big|\Big)^p\Big]^{1/p} \\
&\quad\le \sum_{k=\mathfrak{n}+1}^{n} \gamma_k\, \mathbb{E}\Big[\Big(\mathbb{1}_{\{\mathfrak{N}>k\}} \sum_{r=n_{\ell-1}+1}^{k} \rho_{r-k}\, |X(U_r,\theta_{r-1}) - X(U_r,\theta_{n_{\ell-1}})|\Big)^p\Big]^{1/p} \\
&\quad\le \sum_{k=\mathfrak{n}+1}^{n} \gamma_k \sum_{r=n_{\ell-1}+1}^{k} \varrho_{r-k}\, \mathbb{E}\Big[\Big(\mathbb{1}_{\{\mathfrak{N}>r\}}|X(U_r,\theta_{r-1}) - X(U_r,\theta_{n_{\ell-1}})|\Big)^p\Big]^{1/p} \\
&\quad\le \kappa_3 \tilde{L}(t_{n_\ell} - t_{n_{\ell-1}}) \sum_{k=\mathfrak{n}+1}^{n} \gamma_k.
\end{aligned}
\tag{3.58}
$$

3.) Note that

$$
|\Delta\tilde{\tilde{\Theta}}_{\mathfrak{n},n} - \Delta\tilde{\Theta}_{\mathfrak{n},n}| = |\gamma_n\,(g(\tilde{\tilde{\mathbf{X}}}(n)) - g(\tilde{\mathbf{X}}(n)))| \le \gamma_n\, \|\tilde{\tilde{\mathbf{X}}}(n) - \tilde{\mathbf{X}}(n)\|_{\ell^d_\varrho} \tag{3.59}
$$

and for $n = n_{\ell-1}+1, \dots, n_\ell$,

$$
\begin{aligned}
\|\tilde{\tilde{\mathbf{X}}}(n) - \tilde{\mathbf{X}}(n)\|_{\ell^d_\varrho} &= \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n}|X_k - X(U_{\ell,k},\theta_{n_{\ell-1}})| \\
&\le \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n}|X_k| + \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n}|X(U_{\ell,k},\theta_{n_{\ell-1}})|.
\end{aligned}
\tag{3.60}
$$

Hence,

$$
\begin{aligned}
\mathbb{E}\big[\mathbb{1}_{\{\mathfrak{N}>n_{\ell-1}\}}\|\tilde{\tilde{\mathbf{X}}}(n) - \tilde{\mathbf{X}}(n)\|^p_{\ell^d_\varrho}\big]^{1/p}\big] &\le \|\mathbf{x}\|_{\ell^d_\varrho}\sqrt{\beta}^{\,n-n_0} + 2C \sum_{k=-\infty}^{n_{\ell-1}} \varrho_{k-n} \\
&\le \|\mathbf{x}\|_{\ell^d_\varrho}\sqrt{\beta}^{\,n-n_0} + 2(1-\sqrt{\beta})^{-1} C \varrho_{n_{\ell-1}-n}
\end{aligned}
\tag{3.61}
$$

and

$$
\begin{aligned}
&\mathbb{E}\Big[\mathbb{1}_{\{\mathfrak{N}>n_{\ell-1}\}} \sum_{n=n_{\ell-1}+1}^{n_\ell} |\Delta\tilde{\tilde{\Theta}}_n - \Delta\tilde{\Theta}_n|^p\Big]^{1/p} \\
&\quad\le \gamma_{n_{\ell-1}+1} \sum_{n=n_{\ell-1}+1}^{n_\ell} \big(\|\mathbf{x}\|_{\ell^d_\varrho}\sqrt{\beta}^{\,n-n_0} + 2(1-\sqrt{\beta})^{-1} C \varrho_{n_{\ell-1}-n}\big) \\
&\quad\le \frac{1}{1-\sqrt{\beta}}\gamma_{n_{\ell-1}+1}\sqrt{\beta}^{\,n_{\ell-1}-n_0}\|\mathbf{x}\|_{\ell^d_\varrho} \\
&\qquad + \frac{2}{1-\sqrt{\beta}}\gamma_{n_{\ell-1}+1}\|\varrho\|_{\ell_1} C.
\end{aligned}
\tag{3.62}
$$

4.) The inequalities (IV.a) and (IV.b) are direct consequences of Lemmas 3.1.1 and 3.1.2 when conditioning on $\mathcal{F}_{n_{\ell-1}}$. □

3.2.2 Proof of Lemma 3.2.4

Proof. (i): This is an immediate consequence of

$$
t_{n_\ell} - t_{n_{\ell-1}} \ge \rho\sqrt{\gamma_{n_{\ell-1}+1}} - \gamma_{n_{\ell-1}+1} = (\rho - \sqrt{\gamma_{n_{\ell-1}+1}})\sqrt{\gamma_{n_{\ell-1}+1}}. \tag{3.63}
$$

(ii): Denote by $f : [0,\infty) \to (0,\infty)$ the function satisfying

$$f(t_n) = \gamma_{n+1} \tag{3.64}$$

for every $n \in \mathbb{N}_0$ and that is piecewise linear in between the points of $\{t_0, t_1, \ldots\}$. First note that on each interval (t_{n-1}, t_n)

$$-f'(s) = \frac{\gamma_n - \gamma_{n+1}}{\gamma_n} \leq \zeta\gamma_n \tag{3.65}$$

so that, for $\ell \in \mathbb{N}$,

$$\gamma_{n_\ell+1} \geq \gamma_{n_{\ell-1}+1} - \zeta\gamma_{n_{\ell-1}+1}\rho\sqrt{\gamma_{n_{\ell-1}+1}} \geq (1 - \zeta\rho\sqrt{\gamma_{n_0+1}})\gamma_{n_{\ell-1}+1} \tag{3.66}$$

and

$$\frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_\ell+1}} \leq K. \tag{3.67}$$

One has

$$\gamma_{n_{\ell-1}+1} - \gamma_{n_\ell+1} = f(t_{n_{\ell-1}}) - f(t_{n_\ell}) \leq \zeta\gamma_{n_{\ell-1}+1}(t_{n_\ell} - t_{n_{\ell-1}}) \tag{3.68}$$

and

$$\frac{\gamma_{n_{\ell-1}+1} - \gamma_{n_\ell+1}}{\gamma_{n_\ell+1}} \leq \zeta K(t_{n_\ell} - t_{n_{\ell-1}}). \tag{3.69}$$

□

3.3 ODE based error analysis

In Section 3.2 we cut the training time into blocks

$$n_0 < n_1 < n_2 < \cdots, \qquad h_\ell := t_{n_\ell} - t_{n_{\ell-1}} \asymp \sqrt{\gamma_{n_{\ell-1}+1}},$$

and froze the parameter in the ℓ-th block at $\theta_{n_{\ell-1}}$. The three approximations give the block-level picture

$$\theta_{n_\ell} - \theta_{n_{\ell-1}} \approx h_\ell f(\theta_{n_{\ell-1}}) + \text{ error}$$

Consider ODE:

$$\dot{\Psi}_t = f(\Psi_t). \tag{3.70}$$

And we have Error

$$E_{\ell-1} := \theta_{n_{\ell-1}} - \Psi_{t_{n_{\ell-1}}} \tag{3.71}$$

We wish to have the form which is exactly **the same error structure of Euler method and ODE solution**:

$$E_\ell = E_{\ell-1} + h_\ell(f(\theta_{n_{\ell-1}}) - f(\Psi_{t_{n_{\ell-1}}})) + \text{ error} \tag{3.72}$$

More exactly, we can write

$$E_\ell = \underbrace{E_{\ell-1} + h_\ell(f(\theta_{n_{\ell-1}}) - f(\Psi_{t_{n_{\ell-1}}})) + A_\ell - \tilde{A}_\ell}_{\Upsilon_\ell\text{: predictable part}} + \underbrace{M_\ell}_{\text{centering random part}} \tag{3.73}$$

Then, we want to find what each term means:

$\tilde{A}_\ell$:

This is the error from ODE Euler approximation:

$$\tilde{A}_\ell = \Psi_{t_{n_\ell}} - \left(\Psi_{t_{n_{\ell-1}}} + (t_{n_\ell} - t_{n_{\ell-1}}) f(\Psi_{t_{n_{\ell-1}}})\right) \tag{3.74}$$

A_ℓ:

Define

$$Z_\ell = \theta_{n_\ell \wedge \mathfrak{N}} - \theta_{n_{\ell-1}} - \tilde{\tilde{\Theta}}_{n_{\ell-1}:(n_\ell \wedge \mathfrak{N})}. \tag{3.75}$$

3.75 is the error between the true Adam increment and the stationary frozen increment, which contains error between *Real* Adam and g-Adam, moving parameter and frozen parameter, real old memory and stationdary frozen memory:

$$A_\ell = \mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} \mathbb{E}\left[Z_\ell \middle| \mathcal{F}_{n_{\ell-1}}\right] \tag{3.76}$$

M_ℓ:

Coming from **stationary Adam fluctuation** and **centered approximation fluctuation**. Also we have:

$$\mathbb{E}\big[M_\ell \big| \mathcal{F}_{n_{\ell-1}}\big] = 0$$

Thus, the total error:

$$\|E_\ell\|^p = \|\Upsilon_\ell + M_\ell\|^p \tag{3.77}$$

Recall *Taylor inequality* in 4.1,

Indeed, for $x, y \in \mathbb{R}^d$ and $p \geq 2$,

$$\|x+y\|^p \leq \|x\|^p + p\|x\|^{p-2}\langle x, y\rangle + \frac{p(p-1)}{2}\left(\|x\|^{p-2} + \|y\|^{p-2}\right)\|y\|^2.$$

We apply this conditionally with $x = \Upsilon_\ell$ and $y = M_\ell$. The linear term vanishes by (3.92). Hölder's inequality and the convexity of $z \mapsto z^{p/2}$ then give

$$e_\ell \leq \gamma_{n_\ell+1}^{-1}\left(\mathbb{E}[\|\Upsilon_\ell\|^p]^{2/p} + \mathfrak{p}\,\mathbb{E}[\|M_\ell\|^p]^{2/p}\right), \tag{3.78}$$

where

$$\mathfrak{p} = p - 1 + \left(\frac{p(p-1)}{2}\right)^{2/p}.$$

This formula already decouples the proof into a predictable estimate and a martingale estimate.

Remark 3.3.1. Why do we not apply the triangle inequality directly to $\Upsilon_\ell + M_\ell$? A triangle inequality would make the noise enter at first order through $\|M_\ell\|_{L^p}$. Instead, we use the conditional centering in (3.92). The first-order Taylor term has conditional expectation zero, and the noise enters through $\|M_\ell\|_{L^p}^2$. This is the martingale advantage that would be lost if the bias and the fluctuation were mixed together. This is how **Itô** used in descrete stochastic analysis.

The choice $h_\ell \asymp \sqrt{\gamma}$ balances the two systematic local errors:

$$h_\ell^2 \asymp \gamma.$$

At the same time, the random fluctuation has size

$$\sqrt{\gamma h_\ell} \asymp \gamma^{3/4}.$$

To make the above decomposition precise, let

$$\mathfrak{N} = \inf\{n \geq n_0 : \theta_n \notin V\} \wedge \inf\{n_\ell : \|\theta_{n_\ell} - \Psi_{t_{n_\ell}}\| > \mathfrak{R}_{t_{n_\ell}}\}. \tag{3.79}$$

Here V is the region where the estimates for the innovation are valid, and $\mathfrak{R}_t$ is the radius where the ODE has the local stability which we will use below.

e_ℓ:

Our final goal is

$$\|E_\ell\|_{L^p} = \mathcal{O}(\sqrt{\gamma_{n_\ell+1}}).$$

So we divide the squared error by $\gamma_{n_\ell+1}$ and define

$$e_\ell = \gamma_{n_\ell+1}^{-1}\mathbb{E}\left[\mathbb{1}_{\{\mathfrak{N}\geq n_\ell\}}\|\theta_{n_\ell} - \Psi_{t_{n_\ell}}\|^p\right]^{2/p}. \tag{3.80}$$

Thus, $e_\ell = \mathcal{O}(1)$ means exactly $\|E_\ell\|_{L^p} = \mathcal{O}(\sqrt{\gamma})$.

Now we first estimate the bias A_ℓ. Here "bias" does not mean the bias of a statistical estimator. It means the part of the approximation error which remains after we average the new randomness in this block:

$$Z_\ell = \underbrace{\mathbb{E}[Z_\ell \mid \mathcal{F}_{n_{\ell-1}}]}_{\text{predictable bias}} + \underbrace{Z_\ell - \mathbb{E}[Z_\ell \mid \mathcal{F}_{n_{\ell-1}}]}_{\text{centered fluctuation}}.$$

Conditionally on $\mathcal{F}_{n_{\ell-1}}$, the first term has already been determined. The second term has conditional expectation zero.

Where does this bias come from? Recall the first three parts of Proposition 3.2.1.

(I) Real Adam and g-Adam.
The finite-time correction $1-\beta^n$ is dropped when we change the real Adam increment to $\Delta\Theta_k$. This gives the first approximation error.

(II) Moving parameter and frozen parameter.
In the real process, the stochastic gradient uses θ_{k-1}. In the frozen process, we force all the gradients in the block to use $\theta_{n_{\ell-1}}$. This gives the freezing error of order h_ℓ^2.

(III) Real old memory and stationary frozen memory.
We replace the history before $n_{\ell-1}$ by a stationary history generated at $\theta_{n_{\ell-1}}$. This gives the memory replacement error. The part coming from the initial memory decays with β.

Using Proposition 3.2.1 (I)–(III), we have

$$\|Z_\ell\|_{L^p} \leq \underbrace{\left(\kappa_1\beta^{\frac{1}{2}(n_{\ell-1}+1)-n_0}\|\mathbf{x}\|_{\ell_\varrho^d} + \kappa_2 C\right)\gamma_{n_{\ell-1}+1}\beta^{n_{\ell-1}+1}}_{\text{finite-time correction}} \tag{3.81}$$

$$+ \underbrace{\kappa_3\tilde{L}h_\ell^2}_{\text{freezing parameter}} + \underbrace{\gamma_{n_{\ell-1}+1}\left(\kappa_4 C + \kappa_5\beta^{(n_{\ell-1}-n_0)/2}\|\mathbf{x}\|_{\ell_\varrho^d}\right)}_{\text{history replacement}}. \tag{3.82}$$

By the definition of the ρ-partition,

$$h_\ell^2 \leq \rho^2\gamma_{n_{\ell-1}+1}.$$

So we collect the three sources in

$$\aleph_\ell = \kappa_3\rho^2\tilde{L} + (\kappa_2+\kappa_4)C + (\kappa_1+\kappa_5)\beta^{(n_{\ell-1}-n_0)/2}\|\mathbf{x}\|_{\ell_\varrho^d}, \tag{3.83}$$

and obtain

$$\|Z_\ell\|_{L^p} \le \aleph_\ell \gamma_{n_{\ell-1}+1}. \tag{3.84}$$

Remark 3.3.2. The three errors in (3.81) are not literally deterministic random variables. We call their conditional mean the bias. If these approximation errors still fluctuate around their conditional mean, the centered fluctuation will be put into M_ℓ.

By conditional Jensen,

$$\|A_\ell\|_{L^p} \le \aleph_\ell \gamma_{n_{\ell-1}+1}. \tag{3.85}$$

This is the bias from the Adam approximations. We still have the ODE bias $\tilde{A}_\ell$. Since f is L-Lipschitz and bounded by C' along the ODE,

$$\|\tilde{A}_\ell\| \le \int_{t_{n_{\ell-1}}}^{t_{n_\ell}} \|f(\Psi_s) - f(\Psi_{t_{n_{\ell-1}}})\| \,\mathrm{d}s \tag{3.86}$$

$$\le L \int_{t_{n_{\ell-1}}}^{t_{n_\ell}} \int_{t_{n_{\ell-1}}}^{s} \|f(\Psi_u)\| \,\mathrm{d}u \,\mathrm{d}s \le \frac{1}{2} L C' h_\ell^2. \tag{3.87}$$

We want to put A_ℓ and $\tilde{A}_\ell$ into the same scale. Suppose

$$\gamma_{n_0+1} \le \delta_1 < \rho^2.$$

Lemma 3.2.4 gives

$$\sqrt{\gamma_{n_{\ell-1}+1}} \le \frac{h_\ell}{\rho - \sqrt{\delta_1}}.$$

Thus, define

$$a_\ell = \frac{\aleph_\ell}{\rho - \sqrt{\delta_1}} + \frac{1}{2} L C' \rho. \tag{3.88}$$

Then (3.85) and (3.86) give

$$\|A_\ell - \tilde{A}_\ell\|_{L^p} \le a_\ell \sqrt{\gamma_{n_{\ell-1}+1}}\, h_\ell. \tag{3.89}$$

> a_ℓ:
>
> This is the coefficient of the total predictable bias:
>
> $$\underbrace{A_\ell}_{\text{Adam approximation bias}} - \underbrace{\tilde{A}_\ell}_{\text{ODE Euler bias}}.$$
>
> It will interact with the old error $E_{\ell-1}$ in the squared-error estimate.

Next, we estimate the centered random part. More exactly, define

$$M_\ell := \underbrace{\mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} \big(\tilde{\tilde{\Theta}}_{n_{\ell-1}:n_\ell} - h_\ell f(\theta_{n_{\ell-1}})\big)}_{\text{stationary Adam fluctuation}} \tag{3.90}$$

$$+ \underbrace{\mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} Z_\ell - A_\ell}_{\text{centered approximation fluctuation}}. \tag{3.91}$$

Both terms are centered conditionally on the beginning of the block. Hence

$$\mathbb{E}[M_\ell \mid \mathcal{F}_{n_{\ell-1}}] = 0. \tag{3.92}$$

This is the equality which kills the first-order term in the Taylor inequality above.

For the stationary Adam fluctuation, Proposition 3.2.1 (IV.a) gives

$$\left\| \mathbb{1}_{\{\mathfrak{N} > n_{\ell-1}\}} \left(\tilde{\tilde{\Theta}}_{n_{\ell-1}:n_\ell} - h_\ell f(\theta_{n_{\ell-1}}) \right) \right\|_{L^p} \tag{3.93}$$

$$\le \kappa_6 \tilde{C} \left(\sum_{k=n_{\ell-1}+1}^{n_\ell} \gamma_k^2 \right)^{1/2} \le \kappa_6 \tilde{C} \sqrt{\gamma_{n_{\ell-1}+1} h_\ell}. \tag{3.94}$$

For the centered approximation fluctuation,

$$\| Z_\ell - \mathbb{E}[Z_\ell \mid \mathcal{F}_{n_{\ell-1}}] \|_{L^p} \le 2 \| Z_\ell \|_{L^p}.$$

Thus,

$$\| M_\ell \|_{L^p} \le \kappa_6 \tilde{C} \sqrt{\gamma_{n_{\ell-1}+1} h_\ell} + 2 \aleph_\ell \gamma_{n_{\ell-1}+1}. \tag{3.95}$$

The noise is normalized by $\gamma_{n_\ell+1}$, but the estimate above uses $\gamma_{n_{\ell-1}+1}$. So we need to compare the step sizes at the two ends of the block. Suppose

$$\frac{\gamma_n - \gamma_{n+1}}{\gamma_n^2} \le \zeta, \qquad n \ge n_0 + 1, \qquad \zeta \rho \sqrt{\gamma_{n_0+1}} < 1,$$

and define

$$K = (1 - \zeta \rho \sqrt{\gamma_{n_0+1}})^{-1}.$$

Lemma 3.2.4 gives

$$\frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_\ell+1}} \le K, \qquad \frac{\gamma_{n_{\ell-1}+1}}{\gamma_{n_\ell+1}} \le 1 + \zeta K h_\ell. \tag{3.96}$$

Using $(x+y)^2 \le 2x^2 + 2y^2$, we obtain

$$\gamma_{n_\ell+1}^{-1} \| M_\ell \|_{L^p}^2 \le \left(2K \kappa_6^2 \tilde{C}^2 + \frac{4K}{\rho - \sqrt{\delta_1}} \aleph_\ell^2 \sqrt{\gamma_{n_{\ell-1}+1}} \right) h_\ell. \tag{3.97}$$

We define

$$b_\ell = \mathfrak{p} \left(2K \kappa_6^2 \tilde{C}^2 + \frac{4K}{\rho - \sqrt{\delta_1}} \aleph_\ell^2 \sqrt{\gamma_{n_{\ell-1}+1}} \right). \tag{3.98}$$

> b_ℓ:
>
> This is the coefficient of the martingale variance. Indeed,
>
> $$\| M_\ell \|_{L^p}^2 \sim \gamma h_\ell.$$
>
> After division by γ, the random error enters the normalized recursion with size $b_\ell h_\ell$.

It remains to estimate the predictable part Υ_ℓ. Write

$$\Delta f_{\ell-1} = f(\theta_{n_{\ell-1}}) - f(\Psi_{t_{n_{\ell-1}}}).$$

Suppose the ODE has local monotonicity

$$\langle f(x) - f(\Psi_t), x - \Psi_t \rangle \le -c_1 \|x - \Psi_t\|^2 \tag{3.99}$$

for x in the stopped region. Together with the L-Lipschitz continuity of f, we have

$$\|E_{\ell-1} + h_\ell \Delta f_{\ell-1}\|^2 = \|E_{\ell-1}\|^2 + 2h_\ell \langle E_{\ell-1}, \Delta f_{\ell-1}\rangle + h_\ell^2 \|\Delta f_{\ell-1}\|^2 \tag{3.100}$$
$$\le (1 - 2c_1 h_\ell + L^2 h_\ell^2)\|E_{\ell-1}\|^2. \tag{3.101}$$

In (3.100),

$$\underbrace{-2c_1 h_\ell}_{\text{ODE contraction}} + \underbrace{L^2 h_\ell^2}_{\text{Euler loss}}.$$

The ODE gives the negative part. The explicit Euler approximation loses a second-order term.

There is one more loss. The old error is normalized by $\gamma_{n_{\ell-1}+1}$, while the new error is normalized by $\gamma_{n_\ell+1}$. The second estimate in (3.96) gives the extra factor $1 + \zeta K h_\ell$.

Suppose there exists $c' > 0$ such that

$$2c_1 - L^2 \rho \sqrt{\gamma_{n_0+1}} - \zeta K \ge 2c'. \tag{3.102}$$

Then

$$(1 + \zeta K h_\ell)(1 - 2c_1 h_\ell + L^2 h_\ell^2) \le 1 - 2c' h_\ell. \tag{3.103}$$

Thus, c' is the contraction which remains after subtracting the Euler loss and the step-size normalization loss from the ODE contraction.

Now add the bias $A_\ell - \tilde{A}_\ell$. Suppose

$$\rho a_1 \sqrt{\gamma_{n_0+1}} \le \delta_2. \tag{3.104}$$

Since $a_\ell \le a_1$ and $h_\ell \le \rho\sqrt{\gamma_{n_{\ell-1}+1}}$, we have $a_\ell h_\ell \le \delta_2$. From (3.100) and (3.89),

$$\mathbb{E}[\|\Upsilon_\ell\|^p]^{2/p} \le (1 - 2c_1 h_\ell + L^2 h_\ell^2)\gamma_{n_{\ell-1}+1} e_{\ell-1} \tag{3.105}$$
$$+ \gamma_{n_{\ell-1}+1}(2\sqrt{e_{\ell-1}} + \delta_2) a_\ell h_\ell. \tag{3.106}$$

The second line comes from

$$\|E + (A - \tilde{A})\|^2 = \|E\|^2 + 2\langle E, A - \tilde{A}\rangle + \|A - \tilde{A}\|^2.$$

After dividing by $\gamma_{n_\ell+1}$ and using (3.96) and (3.103),

$$\gamma_{n_\ell+1}^{-1}\mathbb{E}[\|\Upsilon_\ell\|^p]^{2/p} \le (1 - 2c' h_\ell) e_{\ell-1} + K(2\sqrt{e_{\ell-1}} + \delta_2) a_\ell h_\ell. \tag{3.107}$$

Finally, plug (3.107) and (3.97) into the Taylor estimate (3.78). We obtain

$$e_\ell \le (1 - 2c' h_\ell) e_{\ell-1} + 2K a_\ell \sqrt{e_{\ell-1}} h_\ell + K\delta_2 a_\ell h_\ell + b_\ell h_\ell. \tag{3.108}$$

Now we can read every term in this recursion.

$$(1 - 2c' h_\ell) e_{\ell-1}$$

is the old error after ODE contraction. The constant c' is the contraction left after the Euler and normalization losses.

$$2K a_\ell \sqrt{e_{\ell-1}} h_\ell$$

is the cross term between the old error and the predictable bias. It comes from $2\langle E_{\ell-1}, A_\ell - \tilde{A}_\ell \rangle$.

$$K\delta_2 a_\ell h_\ell$$

is the quadratic bias term. It comes from $\|A_\ell - \tilde{A}_\ell\|^2$ and the smallness condition $a_\ell h_\ell \le \delta_2$.

$$b_\ell h_\ell$$

is the martingale variance injection. Since $\mathbb{E}[M_\ell \mid \mathcal{F}_{n_{\ell-1}}] = 0$, its first-order contribution disappears and only the second-order contribution remains.

Remark 3.3.3. The whole argument keeps using the same structure:

$$\boxed{\text{error} = \underbrace{\text{predictable bias}}_{\text{does not vanish after averaging}} + \underbrace{\text{martingale noise}}_{\text{conditional mean zero}}.}$$

The approximations (I)–(III) determine the size of the bias. The fixed parameter analysis determines the size of the martingale noise. ODE monotonicity then contracts the old error, and the recursion (3.108) combines the three mechanisms.

Chapter 4

Appendix

4.1 Taylor inequality

Theorem 4.1.1 (Taylor inequality)**.** *Let $\|\cdot\|$ be a norm induced by a scalar product $\langle\!\langle \cdot, \cdot \rangle\!\rangle$ on $\mathbb{R}^d$ and let $p \geq 2$. One has for every $x, y \in \mathbb{R}^d$ that*

$$\|x+y\|^p \leq \|x\|^p + p\|x\|^{p-2}\langle\!\langle x, y \rangle\!\rangle + \tfrac{1}{2}p(p-1)(\|x\| \vee \|x+y\|)^{p-2}\|y\|^2. \tag{4.1}$$

Proof. Let $x, y \in \mathbb{R}^d$. Note that the function $h(z) = \|z\|^p$ $(z \in \mathbb{R}^d)$ is twice continuously differentiable with differentials

$$Dh(z)(a) = p\,\|z\|^{p-2}\langle\!\langle z, a \rangle\!\rangle \quad \text{and} \quad D^2h(z)(a,b) = p(p-2)\|z\|^{p-4}\langle\!\langle z, a \rangle\!\rangle\langle\!\langle z, b \rangle\!\rangle + p|z|^{p-2}\langle\!\langle a, b \rangle\!\rangle. \tag{4.2}$$

By Taylor's theorem, we thus have

$$|x+y|^p = |x|^p + p\,|x|^{p-2}\langle x, y \rangle + \int_0^1 D^2h(x+ry)(y,y)(1-r)\,\mathrm{d}r. \tag{4.3}$$

Note that for every $r \in [0,1]$, one has $|x+ry| \leq |x| \vee |x+y|$ and

$$\begin{aligned} \|D^2h(x+ry)(y,y)\| &\leq p(p-2)\|x+ry\|^{p-4}\|\langle\!\langle x+ry, y \rangle\!\rangle\langle\!\langle x+ry, y \rangle\!\rangle| + p\|x+ry\|^{p-2}\langle\!\langle y, y \rangle\!\rangle \\ &\leq p(p-1)(\|x\| \vee \|x+y\|)^{p-2}\|y\|^2. \end{aligned} \tag{4.4}$$

The statement follows since $\int_0^1 (1-r)\,\mathrm{d}r = \frac{1}{2}$. □